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BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination : Semester Midterm — Semester: Fall 2025
Duration: **1 Hour 20 Minutes** — Full Marks: **50**

CSE421 / EEE465 : Computer Networks

Answer **ALL** questions. (**Pages: 2**)

Figures in the right margin indicate marks.

Name:	**ID:**	**Section:**

Q1 [CO1]	A host can successfully communicate with another device on the same LAN (ARP reply received, frames exchanged), but cannot reach a web server on a different network. **I.** Which OSI addressing level is confirmed to be working? **II.** Which OSI addressing level is likely misconfigured or failing?	**4**
Q2 [CO2]	A university uses a web proxy cache to serve frequently accessed web objects. At 10:58, the content on the origin server is updated by the content team outside the local network, specifically, a new version of /banner.jpg is uploaded directly to the origin server. The proxy inside the university still holds an older cached copy from before 10:58. ● At 10:59, a student attempts to log in by sending POST /login with JSON credentials through the proxy. ● At 11:00, another student requests GET /banner.jpg via the proxy. **I.** When the proxy receives POST /login, should it return a cached response or forward the request to the origin server? Explain briefly. **II.** Because the origin server's /banner.jpg was updated externally at 10:58, what mechanism should the proxy use to avoid serving the stale cached image to users? Briefly explain.	**3 + 3**
Q3 [CO2]	Sumya, a lecturer, sends assignment guidelines to 50 students by email. Each email goes from her laptop to the university's outgoing mail server, and then to the mail servers of the students. Later, 10 students, trying to open the email, notice that the email disappears from the server after they download it to their devices. **I.** Identify the two main email protocols used in this process. **II.** Explain why both protocols are needed for complete email communication. Also, describe what would happen if only one of these protocols were used.	**2 + 2**
Q4 [CO2]	Arman tries to visit library.campus.net, but his browser returns an error saying **"server not found"**. He then manually tries the server's IP address and successfully loads the website. **I.** What does this tell you about where the failure occurred in the name-to-access process, and how does DNS normally map a hostname to an IP address? **II.** Later, Arman inspects the DNS response and notices that the lookup for campus.net returns a hostname related to mail.campus.net. State what type of DNS record he received, and which type of record he should have received to load the website.	**3 + 3**

Q5 [CO2] (4 + 2)

Rehman and his friends are watching a football match using IPTV. Suddenly, he remembers that he needs to submit his assignment via email. He immediately opened his Outlook App and sent the email containing his large assignment file.

- **I.** If the email sending were to use the same protocol as the live streaming, describe how this would affect the email transmission, and explain why it would be problematic.
- **II.** Some of Rehman's friends notice that the live stream starts lagging while he is uploading the assignment. Briefly explain why uploading a large file can cause the live IPTV stream to lag.

Q6 [CO3] (2 + 2 + 2)

Consider a TCP connection using slow start and congestion avoidance.

- The initial slow-start threshold (ssthresh) is 6 MSS.
- The congestion window (cwnd) changes over the rounds as shown:

Round	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**
cwnd (MSS)	1	2	4	6	7	8	9→?	2	4	5	6	7	8	9

- **I.** Identify why **cwnd = 6** at round 3.
- **II.** What will be the **cwnd** at the end of round 6?
- **III.** Between **Round 4 and 6**, which one indicates more severe congestion?

Q7 (3 + 3 + 3)

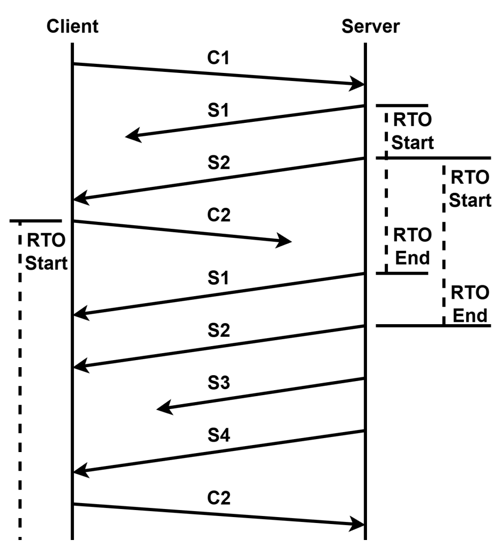
Mubin loads an HTML page in Google Chrome over a **non-persistent HTTP connection** (TCP termination time and the third phase of handshaking are negligible). The page contains **15 objects**, **excluding** the base file. Each object size is **3 MB.**

The **total round-trip time** to load the page is **1280 ms.** The **TCP RTT** takes **40 ms** less than the **per-object HTTP request-and-response RTT**. The server's bandwidth is **32 Mbps.**

- **[CO3] I.** Calculate the **TCP RTT** and the **per-object HTTP request response RTT**.
- **[CO3] II.** Calculate the **total page-load time**.
- **[CO2] III.** Suppose the RTT becomes larger due to network congestion. Explain which part of the page-load time will increase, and briefly explain why.

Q8 (3 + 4 + 2)

At a certain point, the client transmits segment **C1** with a sequence number of **2026** and an acknowledgment number of **1993**. The client's rwnd is **12000** bytes, and the server's rwnd is **10000** bytes right before C1 was sent. The data sizes of the segments C1, C2, S1, S2, S3, S4 are **2500**, **1500**, **500**, **1000**, **3500**, and **3000** bytes, respectively.



- **[CO3] I.** **Determine** whether Selective Repeat or Go-Back-N is used. Note that when the client sends the **second C2 segment**, the advertised rwnd is 7500 bytes.
- **[CO3] II.** **Find** the **sequence** and **acknowledgment** numbers of the **S3** segment.
- **[CO2] III.** **Explain** why the server retransmitted both S1 & S2 segments, although S2 reached the client.

=======================THE END=======================